

Encryption and Data Types -- Guided Notes ANSWER KEY

Unit tp-6.14 | CompTIA Network+ N10-009 Obj. FC0-U71 6.4 | N10-008 FC0-U71 6.4

For instructor use / distribute after students complete the notes.

Question 1

Plaintext is data that is _____ and _____ readable. Ciphertext is data that has been _____ and is not readable without the decryption key. Encryption is the process of converting _____ to _____. Decryption is the reverse. The algorithm + key combination is called a _____.

Answer: Unencrypted; human-; encrypted; plaintext; ciphertext; cipher.

Question 2

Symmetric encryption uses the _____ key for both encryption and decryption. It is faster than asymmetric encryption and is used for _____ data at rest. Key examples: AES-256, 3DES. The major challenge with symmetric encryption is _____ distribution -- how do two parties share the key _____ first meeting without someone intercepting it?

Answer: Same; encrypting; key; before.

Question 3

Asymmetric encryption uses a mathematically linked _____ key pair. The _____ key is shared openly; the _____ key is never shared. Data encrypted with the public key can only be decrypted by the _____ key. This solves the key distribution problem. RSA and ECC are common _____ encryption algorithms.

Answer: Key pair (two); public; private; private; asymmetric.

Question 4

In practice, HTTPS uses a _____ of both. Asymmetric encryption is used during the TLS _____ to exchange a one-time _____ key. Then symmetric encryption (AES) encrypts the actual _____ data because symmetric is much _____ than asymmetric for bulk data.

Answer: Hybrid; handshake; session (symmetric); session; faster.

Question 5

Data at _____ is data stored on a drive, database, or backup media. It should be encrypted with AES-256. Data in _____ is data actively moving across a network. It should be encrypted using TLS. Data in _____ is data actively being processed in RAM by a CPU. This is the hardest to encrypt and is a target of _____ scraping malware.

Answer: Rest; transit; use; memory.

Question 6

Data types relevant to security classification: _____ data is personal/identifiable information. _____ data is financial records, trade secrets, IP. _____ data has no identified source and presents lower risk. Data _____ is the process of removing identifying fields so data cannot be linked

to an individual. _____ replaces sensitive fields with random tokens.

Answer: Sensitive (PII/PHI); proprietary (confidential); anonymous; anonymization; tokenization.

Question 7

A common encryption misconception: password _____ is NOT encryption. Hashing is _____ -way -- you cannot recover the original password from the hash. Encryption is _____ -way -- you can recover the plaintext with the key. Passwords should be _____ with a salt. Files should be _____.

Answer: Hashing; one; two (reversible); hashed; encrypted.

Question 8 -- Real World Application

REAL WORLD: A company stores customer credit card numbers in their database in plaintext so customer service reps can read them when handling calls. Their database is later breached and all 50,000 card numbers are stolen. Explain what they should have stored instead of plaintext card numbers and describe how tokenization would have made the breach far less damaging.

Answer: Instead of plaintext card numbers, the company should have stored tokens. Tokenization replaces each real credit card number with a randomly generated token (e.g., 4x3a-9y2b-...) that has no mathematical relationship to the original number. The real card number is stored securely in a separate, heavily protected token vault managed by a PCI-compliant tokenization provider. How this limits the breach: when the attacker steals the database, they get 50,000 random tokens -- not credit card numbers. The tokens are useless for making fraudulent purchases because they cannot be reversed to the real card numbers without access to the token vault (which was not breached). Customer service reps see the token in their system and it is sufficient to identify the customer's payment method context, but not to make charges. This is why major payment processors (Stripe, Braintree) use tokenization -- no merchant ever stores real card numbers, so merchant breaches cannot expose cards. The company also likely violated PCI-DSS by storing plaintext card numbers, which mandates encryption or tokenization of stored cardholder data.